

Experiment No - 08

- a) Simulate Capturing and analyzing Ethernet frames.**
- b) Simulate HTTP GET/POST interaction**
- c) Simulate capturing a bulk TCP transfer from your computer to a remote server.**

a) Simulate Capturing and Analyzing Ethernet Frames

Objective

To capture and study Ethernet frames (Layer 2 packets).

Tool

- Wireshark

Steps

1. Open Wireshark
2. Select your active network interface (Wi-Fi/Ethernet)
3. Start capturing packets
4. Perform some network activity (e.g., open a website)
5. Stop capture after a few seconds

Filter Ethernet Frames

- Use filter:

eth

What to Analyze

Click any packet → Expand **Ethernet II** section:

- **Destination MAC Address**
- **Source MAC Address**
- **Type (e.g., IPv4 = 0x0800)**

Result / Conclusion

Can observe how data is encapsulated into Ethernet frames with MAC addressing at Layer 2.

b) Simulate HTTP GET/POST Interaction

Objective

To observe HTTP request-response (GET and POST methods).

Tools

- Wireshark
 - Google Chrome
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Steps for HTTP GET

1. Start capture in Wireshark
2. Open browser and visit a site:

`http://example.com`

3. Stop capture

Filter

`http`

Observe GET

- Method: **GET**
 - Request line:

`GET / HTTP/1.1`
 - Response:
 - Status: 200 OK
 - Content-Type, Content-Length
-

Steps for HTTP POST

1. Start capture
2. Submit a form (e.g., login page or use test API like `httpbin.org`)
3. Stop capture

Observe POST

- Method: **POST**
 - Request includes:
 - Form data (username, password, etc.)
 - Response:
 - Server reply (200 OK / 302 Redirect)
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Conclusion

- GET → retrieves data
- POST → sends data to server

c) Simulate Capturing a Bulk TCP Transfer

Objective

To analyze TCP behavior during large data transfer.

Tools

- Wireshark
 - Browser or file download
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Steps

1. Start Wireshark capture
 2. Download a large file (e.g., from a website or cloud)
 3. Stop capture after download completes
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Filter TCP Traffic

tcp

What to Analyze

1. TCP 3-Way Handshake

- SYN
- SYN-ACK
- ACK

2. Data Transfer

- Sequence numbers
- Acknowledgments
- Window size

3. Flow Control & Congestion

- Retransmissions
- Duplicate ACKs

4. Connection Termination

- FIN / ACK sequence
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Optional Filter (for specific connection)

Right-click a packet →
Follow → TCP Stream

Conclusion

Can observe how TCP ensures:

- Reliable delivery
- Ordered data transfer
- Congestion control